

Hafsah Hussain Ahamed

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Computer Science & Design · SUTD · Software Engineering Track · AI Minor

I build full-stack systems with a focus on AI integration, cloud infrastructure, and DevOps.

EDUCATION

Singapore University of Technology and Design

Sep 2023 to May 2027

- Bachelor of Engineering (Computer Science and Design)
- Relevant Coursework:
 - 50.001 Information Systems & Programming, 50.003 Elements of Software Construction, 50.004 Algorithms, 50.005 Computer System Engineering, 50.007 Machine Learning, 50.043 Database Systems, 50.039 Deep Learning
- Fifth Row Activities: Google Developer Club, Muslim Society

Temasek Polytechnic

Apr 2020 to May 2023

- Diploma in Information Technology
- GPA: 3.69/4.00, Director's List 2021/2022
- Relevant Coursework: Full Stack Web Development, Machine Learning, Data Mining & Business Analytics

WORK EXPERIENCE

Ernst & Young (EY)

Sep 2025 to Jan 2026

Software Engineer Intern

- Developed and enhanced backend API functionalities using ASP.NET C# , IIS, and SQL Server.
- Ensured API–frontend consistency via Swagger documentation and contributed code through version control.

Accenture Pte Ltd

Jun 2022 to Feb 2023

Software Engineer Intern

- Enhanced visualization and analytics dashboards in Qlik Sense and implemented functions in the PEGA system.
- Liaised with users to gather requirements, prepared test plans, and conducted User Acceptance Tests (UAT).
- Deployed fixes to production environment and created or updated documentation (FDS).

ACADEMIC PROJECTS

Diabetic Retinopathy Detection

Jan 2026 – May 2026

50.039 Theory and Practice of Deep Learning

- Developed a deep learning pipeline using an Attention & Residual U-Net (PyTorch) to detect and segment diabetic retinopathy from retinal fundus images
- Benchmarked 6 neural network architectures to identify the best-performing model for lesion segmentation and severity classification
- Built and deployed a Gradio demo with real-time segmentation overlay and severity/confidence scoring

Ascenda Hotel Booking Website

May 2025 – Aug 2025

50.003 Elements of Software Construction

- Developed a MERN full-stack hotel booking platform with real-time Ascenda API integration, advanced filtering, sorting, and smooth navigation.
- Developed responsive UI and polling for incomplete API responses and restructured into microservices for scalability.
- Implemented Jest and Cypress tests across core flows and containerized the app with Docker Compose.

Task Tales – Gamified Productivity App

Jan 2025 – Apr 2025

50.001 Information Systems & Programming

- Created a gamified productivity Java app designed to incentivize good habits in task management through engaging gameplay.
- Handled Git branching and merge conflicts to ensure stable, crash-free updates to the main app branch.
- Collaborated with team to brainstorm features and improve app design, fostering idea sharing and effective teamwork.
- Award : Singtel Honourable Mention Award

LANGUAGES AND TECHNOLOGIES

- **Front-End Development:** Java, Angular, JavaScript, React
- **Back-End Development:** MySQL, Microsoft SQL Server, MongoDB databases, Firebase, Supabase, NodeJS, ASP.Net
- **AI/ML :** OpenCV, Tableau, PowerBI, Knime, QlikSense, Python
- **DevOps & Tools:** Git, AWS, Docker, Jenkins

AWARDS

2025: Singtel Honourable Mention Project Award

2024: Top 10 Design Thinking Project

2023: Edusave Skills Award 2023